

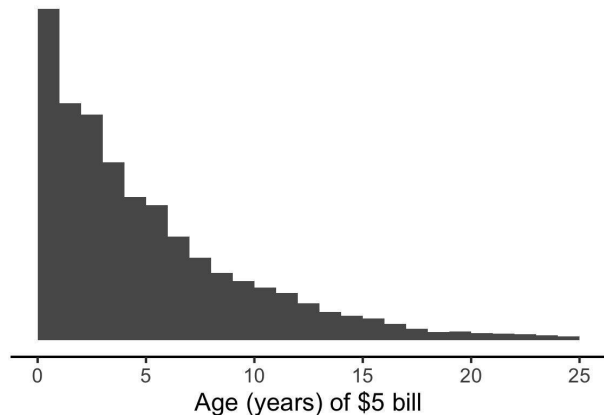
Testing the formulas for SE

MATH 213

Group 1 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

What will the standard error be for this kind of sampling?

What shape will the sampling distribution have?

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

What will the standard error be for this kind of sampling?

What shape will the sampling distribution have?

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud. Put your sampling-distribution histograms here along with the centers and standard errors.

Center for proportion:

SE for proportion:

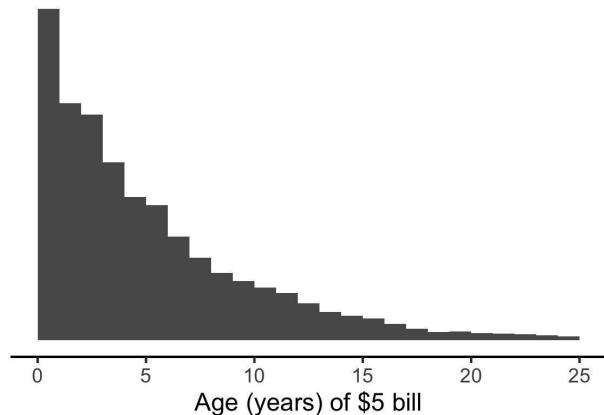
Center for mean:

SE for mean:

Group 2 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be? **5.55 years**

What will the standard error be for this kind of sampling?
 $5.73 / \sqrt{500} = 0.2562$

What shape will the sampling distribution have?
Bell-shaped

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

0.5334

What will the standard error be for this kind of sampling?

0.022

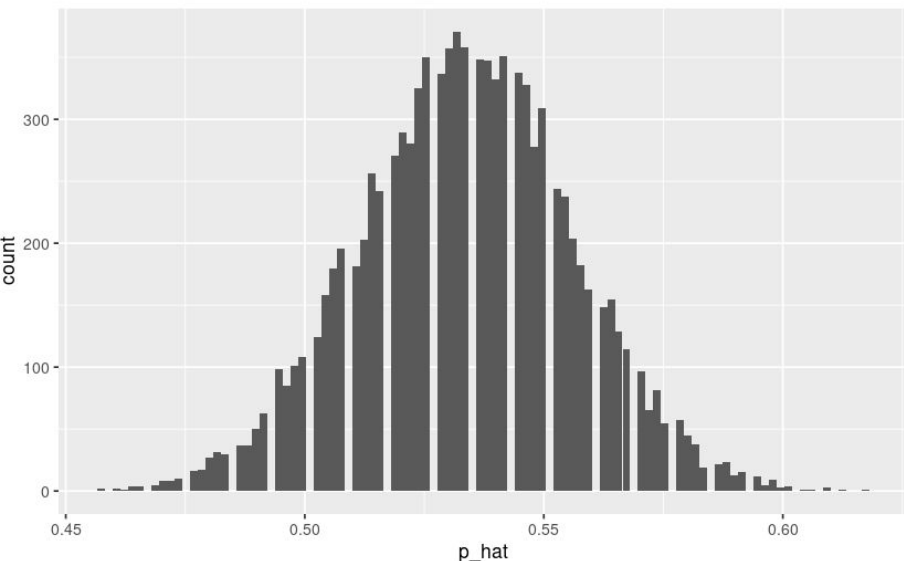
What shape will the sampling distribution have?

Bell-shaped

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud! Put your sampling-distribution histograms here along with the centers and standard errors.

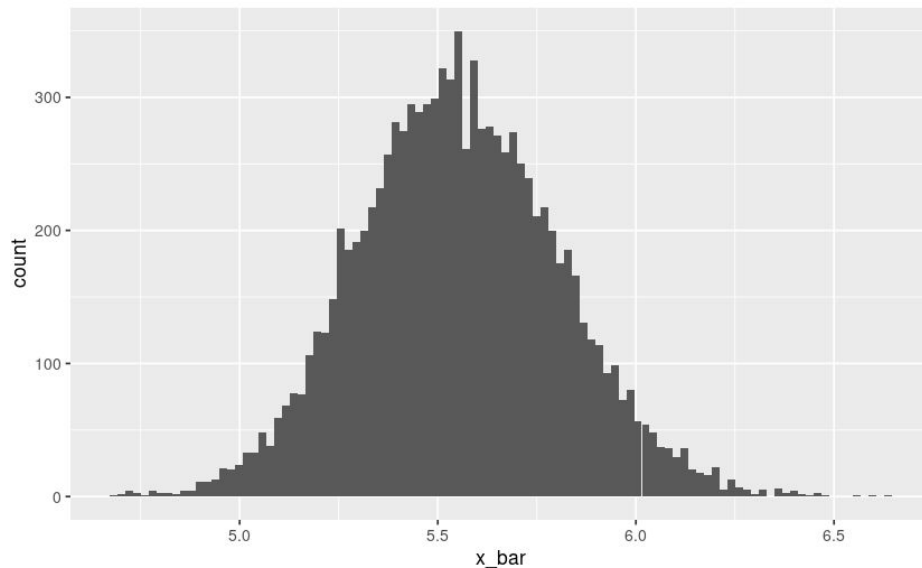
Center for proportion: 0.5336328

SE for proportion: 0.02245834



Center for mean: 5.552702

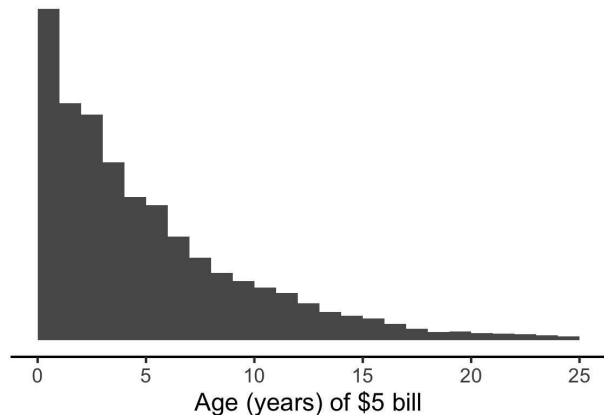
SE for mean: 0.25388



Group 3 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55 years

What will the standard error be for this kind of sampling?

$$5.73/\sqrt{500} = 0.256$$

What shape will the sampling distribution have?
bell-shaped

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?
53.34%

What will the standard error be for this kind of sampling?

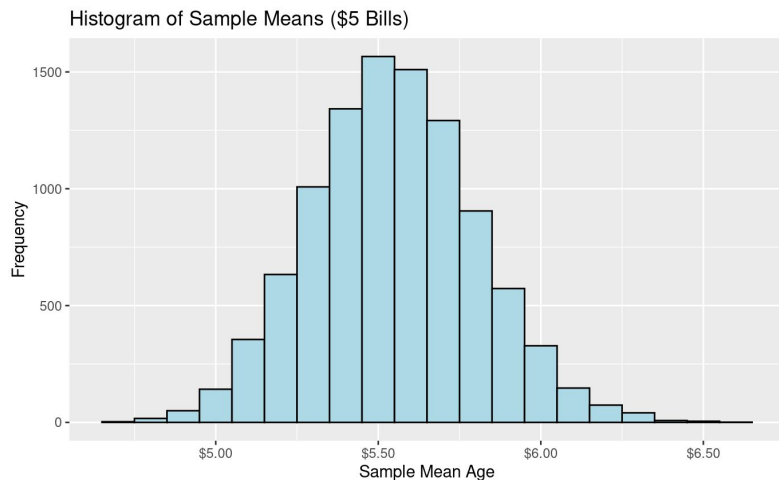
0.022

What shape will the sampling distribution have?
bell-shaped

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud. Put your sampling-distribution histograms here along with the centers and standard errors.

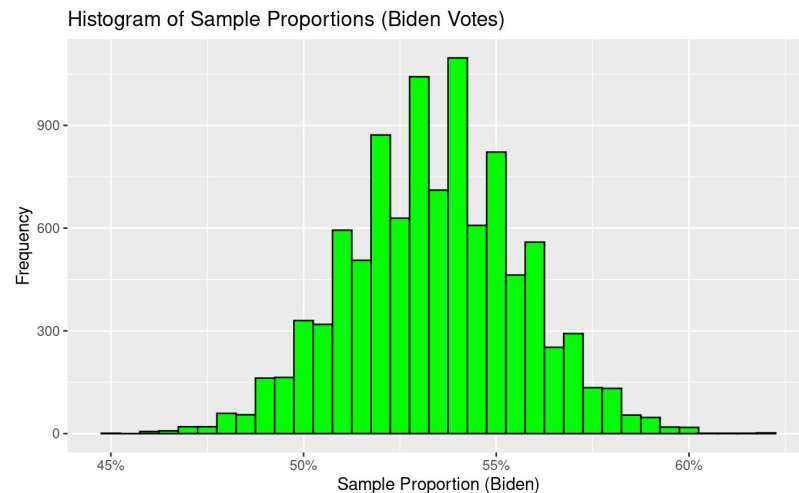
Center for proportion: 5.548034

SE for proportion: 0.2552045



Center for mean: 0.5339952

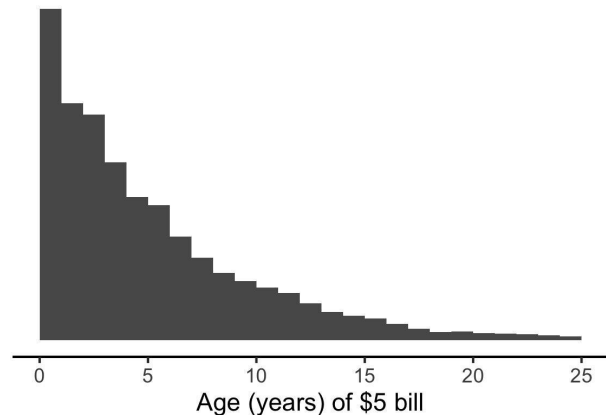
SE for mean: 0.02234204



Group 4 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55

What will the standard error be for this kind of sampling?

.2562533902

What shape will the sampling distribution have?

Bell curve

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

It should be roughly about the population mean of 53.34% but it may be slightly off due to variances in sampling.

What will the standard error be for this kind of sampling?

.02231

What shape will the sampling distribution have?

Bell curve

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud! Put your sampling-distribution histograms here along with the centers and standard errors.

Center for proportion:

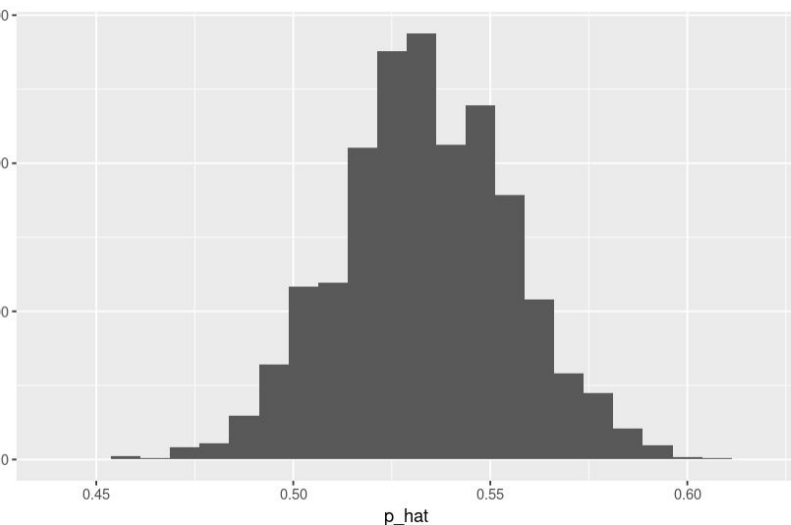
SE for proportion:

mean_phat
<dbl>

0.533487

sd_phat
<dbl>

0.02209883



Center for mean:

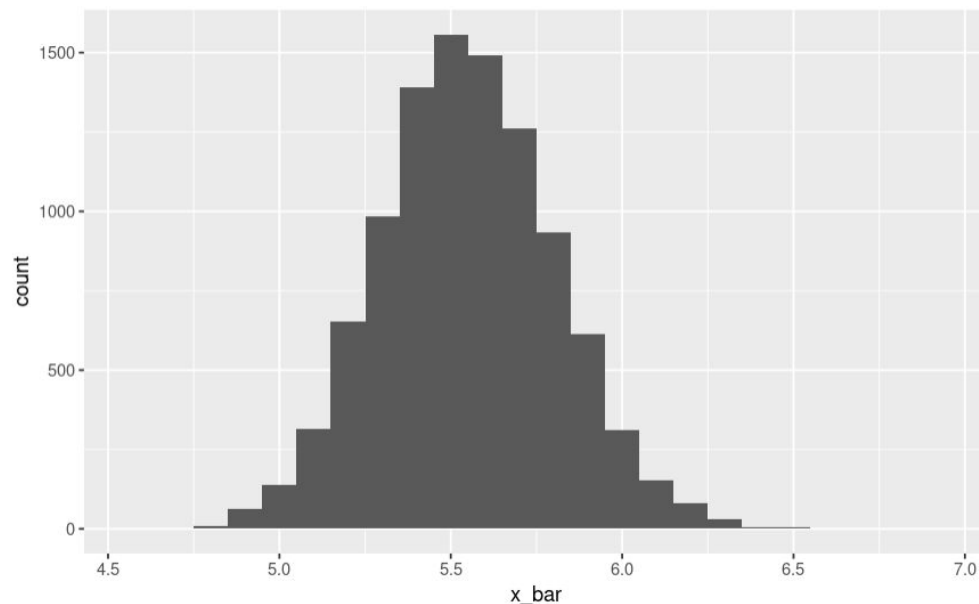
SE for mean:

mean_xbar
<dbl>

5.549124

sd_xbar
<dbl>

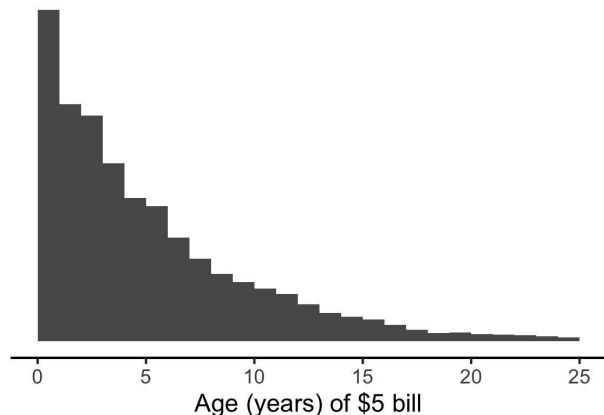
0.2551921



Group 5 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

The center will be at the population parameter which is around 5.55 years

What will the standard error be for this kind of sampling?

standard deviation = 5.73

Sample size = 500

Standard error = $5.73 / \sqrt{500} = 0.25625339022$

What shape will the sampling distribution have?

The shape will be bell shaped

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

It will be at the population parameter which is at 53.34%

What will the standard error be for this kind of sampling?

$\text{sqrt}((0.5334 * (1 - 0.5334)) / 500) = 0.02231073463$

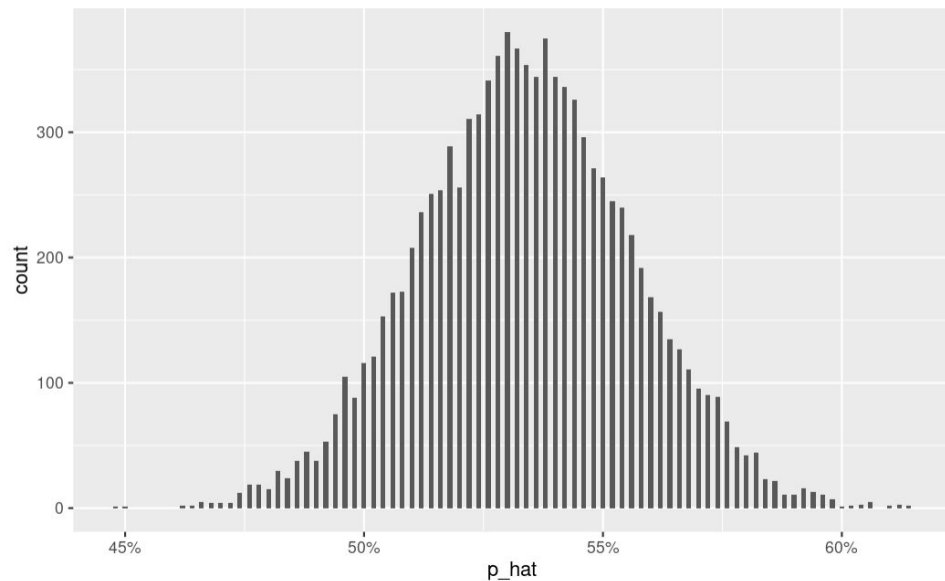
What shape will the sampling distribution have?

Bell Shaped

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud. Put your sampling-distribution histograms here along with the centers and standard errors.

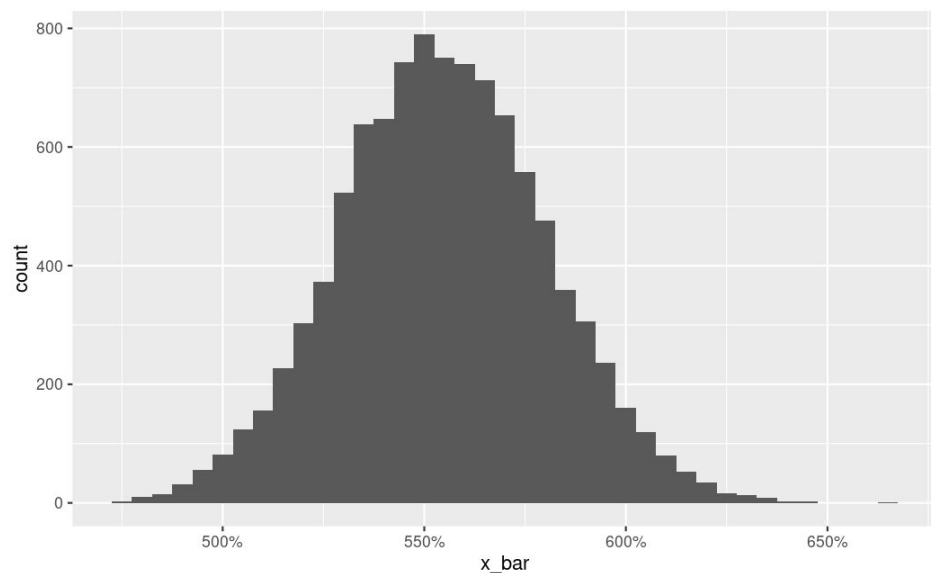
Center for proportion: 0.5338502

SE for proportion:



Center for mean: 5.548156

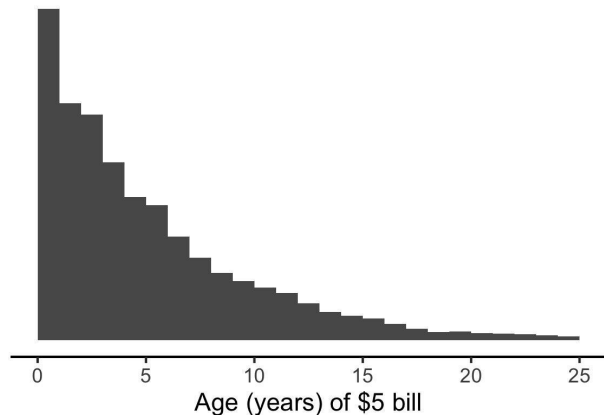
SE for mean:



Group 6 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55

What will the standard error be for this kind of sampling?

$5.73 / \sqrt{500} = 0.2562$

What shape will the sampling distribution have?

Bell curve

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

53.34

What will the standard error be for this kind of sampling?

$\text{sqrt}((.5334(1 - .5334))/500) = 0.02231073463$

What shape will the sampling distribution have?

Bell

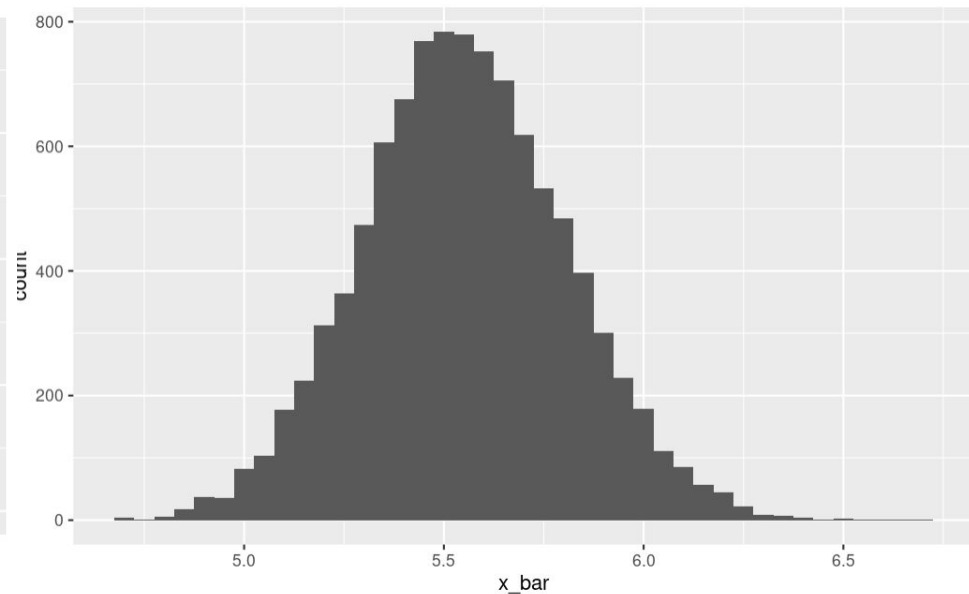
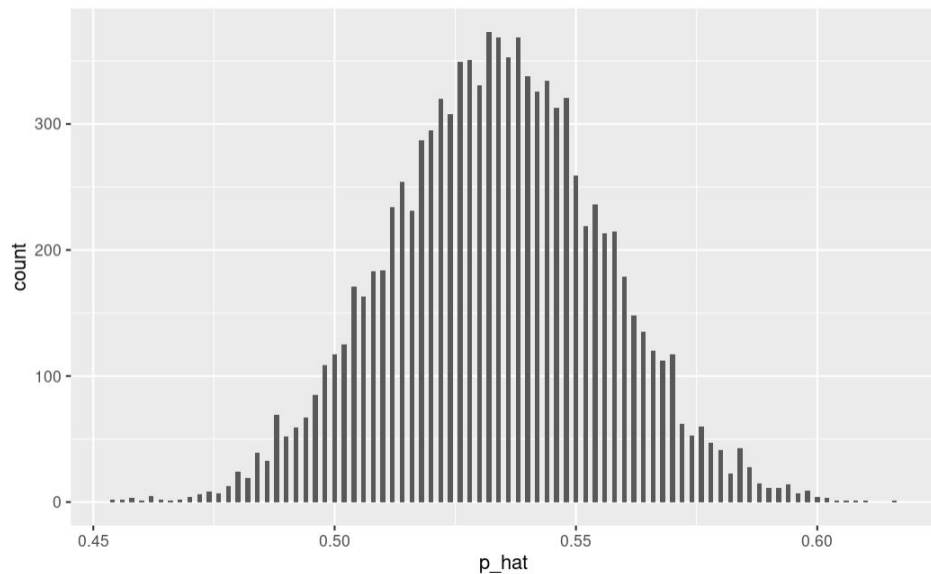
Test your predictions from the previous two examples using the file that I have shared on Posit Cloud! Put your sampling-distribution histograms here along with the centers and standard errors.

Center for proportion: 0.533519

SE for proportion: 0.02223606

Center for mean: 5.552137

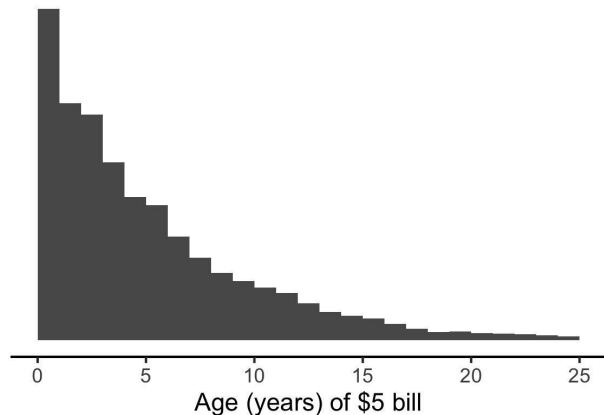
SE for mean: 0.2564854



Group 7 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55

What will the standard error be for this kind of sampling?

$\sigma/\sqrt{n}$

$5.73/\sqrt{500}$

Std er. 0.256

What shape will the sampling distribution have?

Right skewed

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

0.53

What will the standard error be for this kind of sampling?

$\sqrt{p(1-p)/n}$

$\sqrt{0.53(1-0.53)/500}$

0.00068

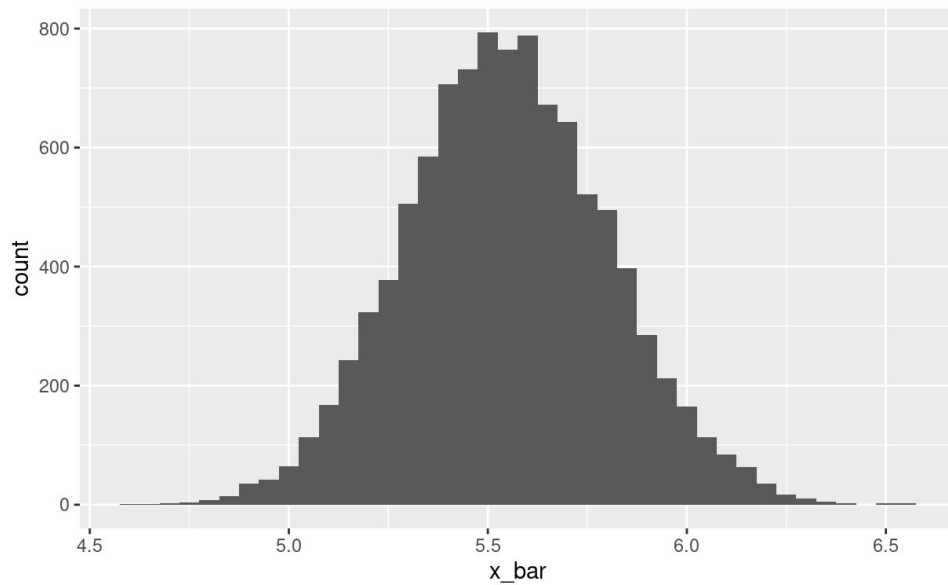
What shape will the sampling distribution have?

Normal bell curve

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud. Put your sampling-distribution histograms here along with the centers and standard errors.

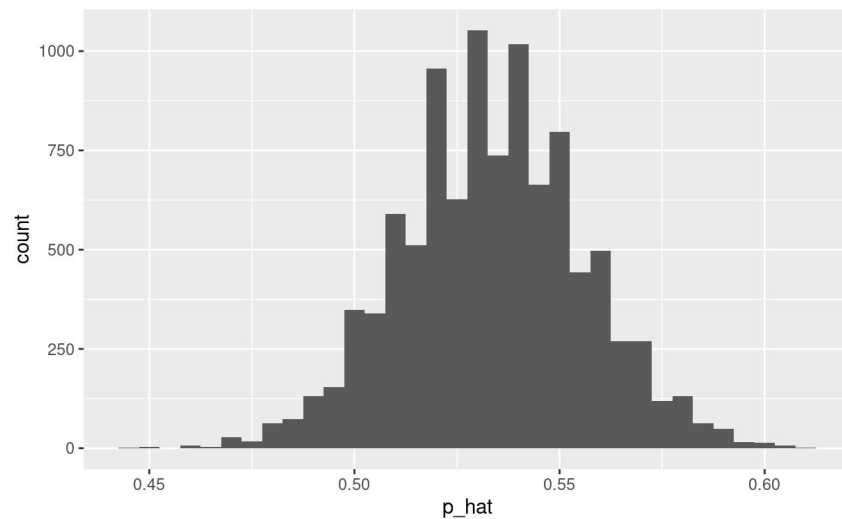
Center for proportion: 5.547856

SE for proportion: 0.2547785



Center for mean: 0.5334876

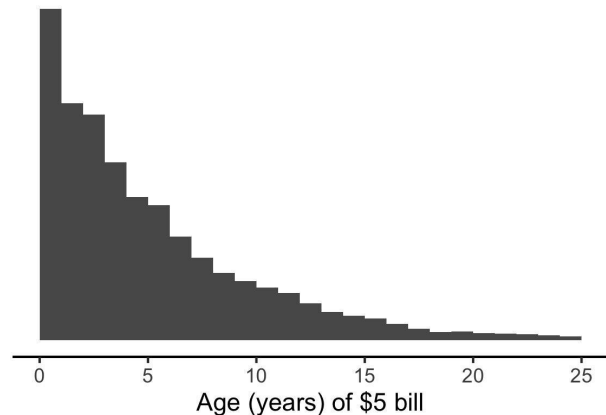
SE for mean: 0.0222921



Group 8 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55

What will the standard error be for this kind of sampling?

.25

What shape will the sampling distribution have?

Bell shaped

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?
53.34

What will the standard error be for this kind of sampling?

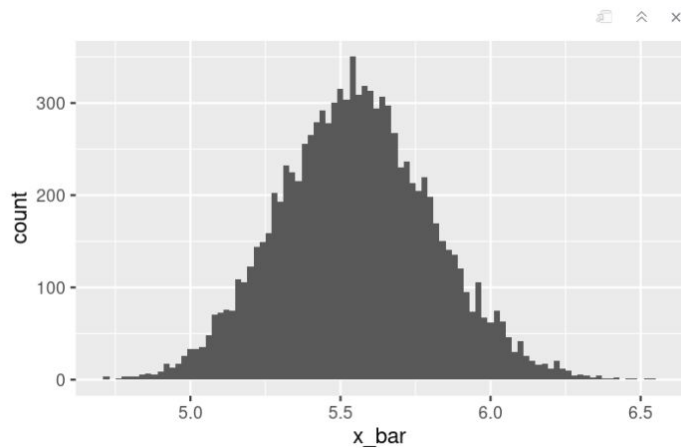
It will be 0.022

What shape will the sampling distribution have?
Bell shaped

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud! Put your sampling-distribution histograms here along with the centers and standard errors.

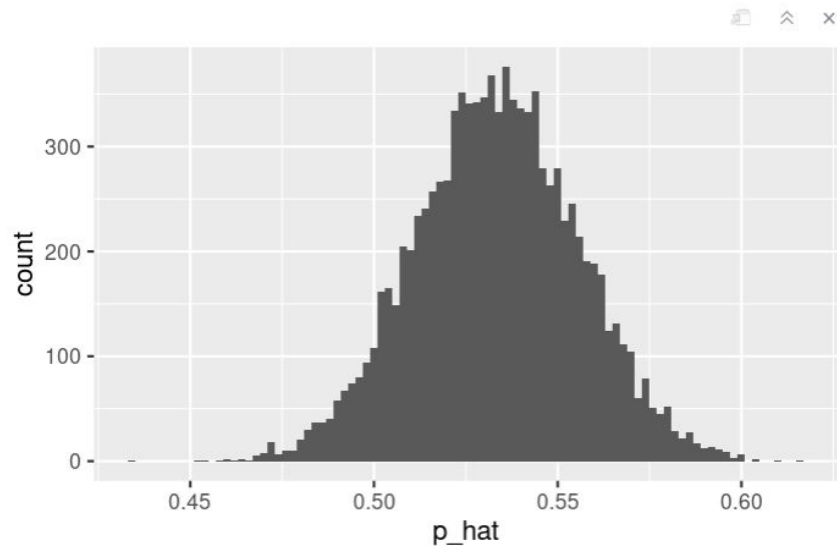
Center for proportion: 5.55

SE for proportion: .255555



Center for mean: .5333

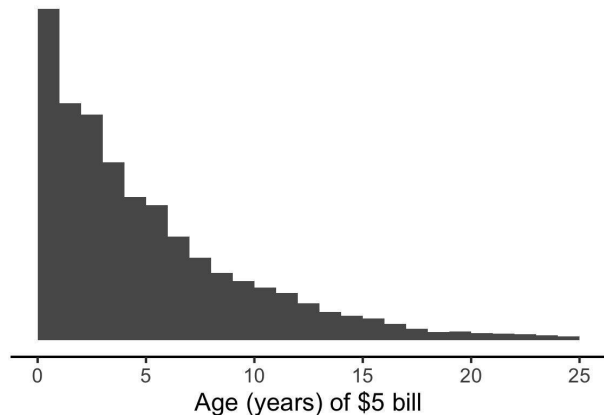
SE for mean: 0.0223



Group 9 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55

What will the standard error be for this kind of sampling?

0.256

What shape will the sampling distribution have?

Normal

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

0.5334

What will the standard error be for this kind of sampling?

0.022

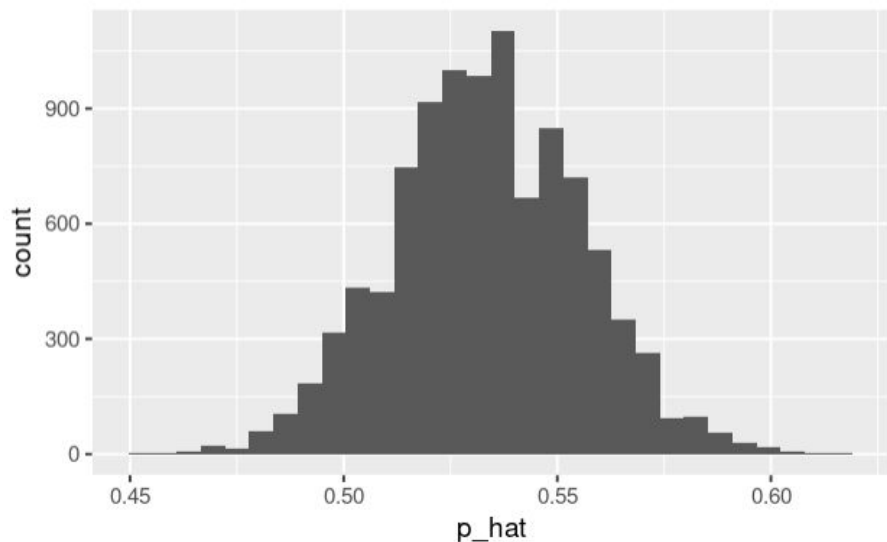
What shape will the sampling distribution have?

Normal

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud. Put your sampling-distribution histograms here along with the centers and standard errors.

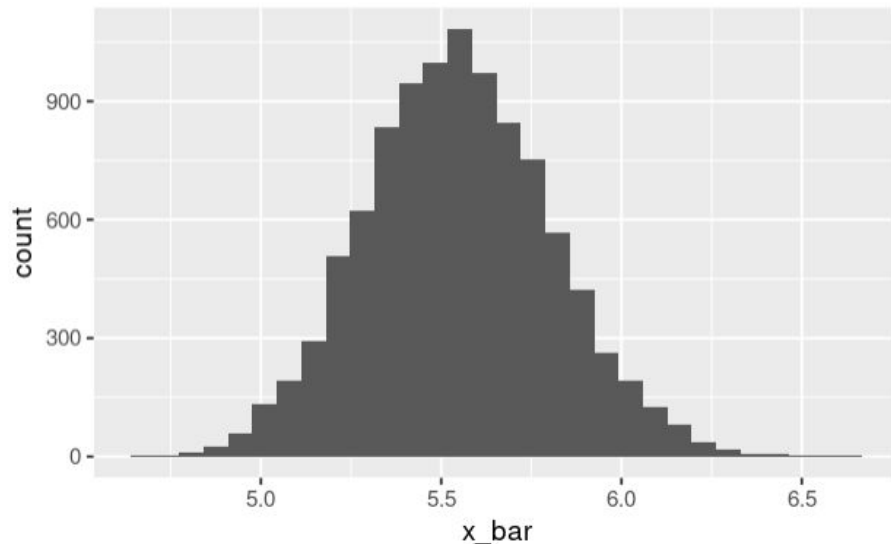
Center for proportion: **0.5335404**

SE for proportion: **0.02222094**



Center for mean: **5.546372**

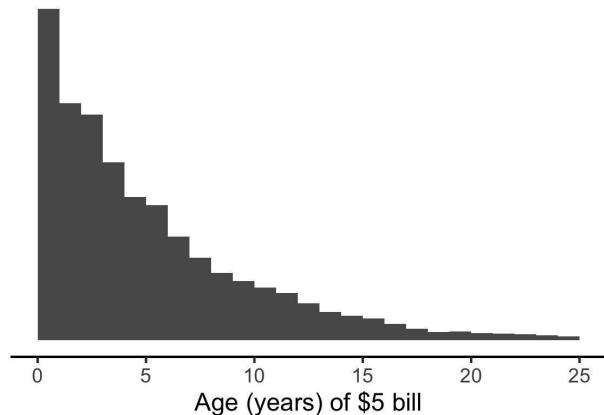
SE for mean: **0.2577619**



Group 10 - Example 1

Suppose we consider the population of \$5 bills in circulation in the United States and we are interested in the ages of the \$5 bills. The mean age of bills in the population is 5.55 years, and the standard deviation in the ages is 5.73 years.

Hypothetical distribution of ages of \$5 bills



Imagine taking samples of size 500 from this population and computing the mean age of bills in each sample.

What will the center of the sampling distribution for the mean age be?

5.55

What will the standard error be for this kind of sampling?

$5.73 / \sqrt{500} = 0.256$

What shape will the sampling distribution have?

normal

Example 2

In the 2020 election, Joe Biden won 53.34% of the vote in Frederick County. Imagine taking samples of 500 Frederick County voters and asking for whom each person would vote. We could then compute proportion of Biden voters in each sample.

What will the center of the sampling distribution for the sample proportion of Biden voters be?

It will be centered around 53.34%

What will the standard error be for this kind of sampling?

$$\text{Sqrt}(0.5334(1-0.5334) / 500) = 0.0223$$

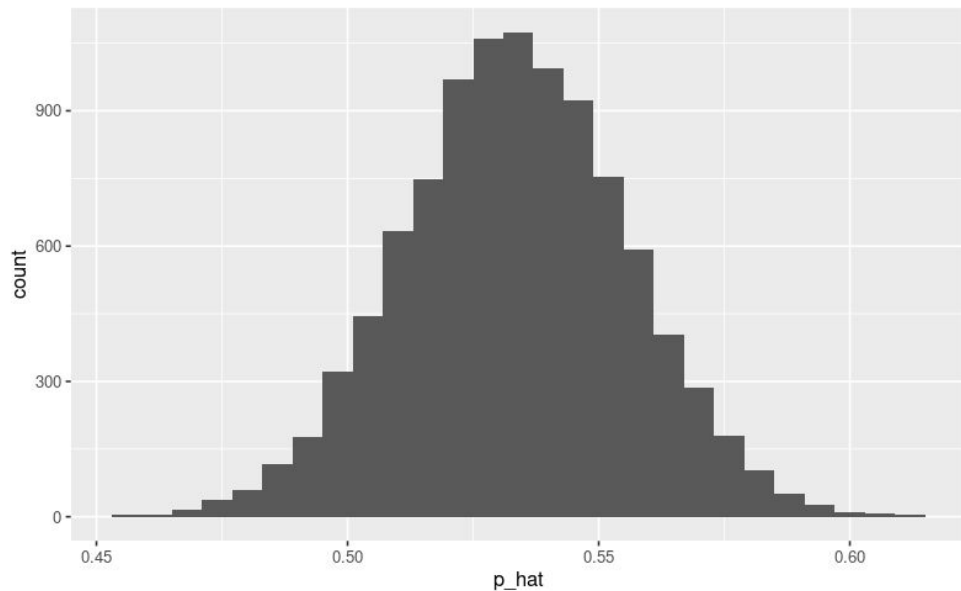
What shape will the sampling distribution have?

Bell shape

Test your predictions from the previous two examples using the file that I have shared on Posit Cloud! Put your sampling-distribution histograms here along with the centers and standard errors.

Center for proportion: 0.5334068

SE for proportion: 0.02245834



Center for mean: 5.55116

SE for mean: 0.25428

